YASKAWA

YRC1000micro OPTIONS
INSTRUCTIONS

FOR SEARCH FUNCTION

Upon receipt of the product and prior to initial operation, read these instructions thoroughly, and retain for future reference.

MOTOMAN INSTRUCTIONS

MOTOMAN-□□□ INSTRUCTIONS
YRC1000micro INSTRUCTIONS
YRC1000micro OPERATOR'S MANUAL
YRC1000micro MAINTENANCE MANUAL
YRC1000micro ALARM CODES (MAJOR ALARMS) (MINOR ALARMS)

The YRC1000micro alarm codes above consists of "MAJOR ALARMS" and "MINOR ALARMS".

Please have the following information available when contacting Yaskawa Customer Support:
- System
- Primary Application
- Software Version *(Located on Programming Pendant by selecting: {Main Menu} - {System Info} - {Version})*
  Robot Serial Number *(Located on robot data plate)*
  Robot Sales Order Number *(Located on controller data plate)*

Part Number: 181287-1CD
Revision: 0

Yaskawa Customer Support
- 24-hour Telephone Number: **(937) 847-3200**
  Use for urgent or emergency needs for technical support, service and/or replacement parts
- Routine Technical Inquiries: *techsupport@motoman.com*
  Allow up to 36 hours for response

MANUAL NO.
HW1484472

DANGER

- This manual explains the search function of the YRC1000micro system. Read this manual carefully and be sure to understand its contents before handling the YRC1000micro. Any matter not described in this manual must be regarded as "prohibited" or "improper".
- General information related to safety are described in "Chapter 1. Safety" of the YRC1000micro INSTRUCTIONS. To ensure correct and safe operation, carefully read "Chapter 1. Safety" of the YRC1000micro INSTRUCTIONS.

CAUTION

- In some drawings in this manual, protective covers or shields are removed to show details. Make sure that all the covers or shields are installed in place before operating this product.
- YASKAWA is not responsible for incidents arising from unauthorized modification of its products. Unauthorized modification voids the product warranty.

NOTICE

- The drawings and photos in this manual are representative examples and differences may exist between them and the delivered product.
- YASKAWA may modify this model without notice when necessary due to product improvements, modifications, or changes in specifications. If such modification is made, the manual number will also be revised.
- If your copy of the manual is damaged or lost, contact a YASKAWA representative to order a new copy. The representatives are listed on the back cover. Be sure to tell the representative the manual number listed on the front cover.

Notes for Safe Operation

Read this manual carefully before installation, operation, maintenance, or inspection of the YRC1000micro.

In this manual, the Notes for Safe Operation are classified as “DANGER”, “WARNING”, “CAUTION”, or “NOTICE”.

DANGER

Indicates an imminently hazardous situation which, if not avoided, will result in death or serious injury. Safety Signs identified by the signal word DANGER should be used sparingly and only for those situations presenting the most serious hazards.

WARNING

Indicates a potentially hazardous situation which, if not avoided, will result in death or serious injury. Hazards identified by the signal word WARNING present a lesser degree of risk of injury or death than those identified by the signal word DANGER.

CAUTION

Indicates a hazardous situation, which if not avoided, could result in minor or moderate injury. It may also be used without the safety alert symbol as an alternative to “NOTICE”.

NOTICE

NOTICE is the preferred signal word to address practices not related to personal injury. The safety alert symbol should not be used with this signal word. As an alternative to “NOTICE”, the word “CAUTION” without the safety alert symbol may be used to indicate a message not related to personal injury.

Even items described as “CAUTION” may result in a serious accident in some situations.
At any rate, be sure to follow these important items.

NOTE

To ensure safe and efficient operation at all times, be sure to follow all instructions, even if not designated as “DANGER”, “WARNING” and “CAUTION".

DANGER

- Before operating the manipulator, make sure the servo power is turned OFF by performing the following operations. When the servo power is turned OFF, the SERVO ON LED on the programming pendant is turned OFF.
  - Press the emergency stop button on the programming pendant or on the external control device, etc.
  - Disconnect the safety plug of the safety fence. (when in the play mode or in the remote mode)

If operation of the manipulator cannot be stopped in an emergency, personal injury and/or equipment damage may result.

Fig. : Emergency Stop Button

- Before releasing the emergency stop, make sure to remove the obstacle or error caused the emergency stop, if any, and then turn the servo power ON.

Failure to observe this instruction may cause unintended movement of the manipulator, which may result in personal injury.

Fig. : Release of Emergency Stop

- Observe the following precautions when performing a teaching operation within the manipulator's operating range:
  - Be sure to perform lockout by putting a lockout device on the safety fence when going into the area enclosed by the safety fence. In addition, the operator of the teaching operation must display the sign that the operation is being performed so that no other person closes the safety fence.
  - View the manipulator from the front whenever possible.
  - Always follow the predetermined operating procedure.
  - Always keep in mind emergency response measures against the manipulator’s unexpected movement toward a person.
  - Ensure a safe place to retreat in case of emergency.

Failure to observe this instruction may cause improper or unintended movement of the manipulator, which may result in personal injury.

- Confirm that no person is present in the manipulator's operating range and that the operator is in a safe location before:
  - Turning ON the YRC1000micro power
  - Moving the manipulator by using the programming pendant
  - Running the system in the check mode
  - Performing automatic operations

Personal injury may result if a person enters the manipulator's operating range during operation. Immediately press an emergency stop button whenever there is a problem. The emergency stop button is located on the right of the programming pendant.

- Read and understand the Explanation of the Warning Labels before operating the manipulator.

DANGER

- In the case of not using the programming pendant, be sure to supply the emergency stop button on the equipment. Then before operating the manipulator, check to be sure that the servo power is turned OFF by pressing the emergency stop button.
  Connect the external emergency stop button to the 4-14 pin and 5-15 pin of the Safety connector (Safety).
- Upon shipment of the YRC1000micro, this signal is connected by a jumper cable in the dummy connector. To use the signal, make sure to supply a new connector, and then input it.

If the signal is input with the jumper cable connected, it does not function, which may result in personal injury or equipment damage.

WARNING

- Perform the following inspection procedures prior to conducting manipulator teaching. If there is any problem, immediately take necessary steps to solve it, such as maintenance and repair.
  - Check for a problem in manipulator movement.
  - Check for damage to insulation and sheathing of external wires.
- Return the programming pendant to a safe place after use.

If the programming pendant is left unattended on the manipulator, on a fixture, or on the floor, etc., the Enable Switch may be activated due to surface irregularities of where it is left, and the servo power may be turned ON. In addition, in case the operation of the manipulator starts, the manipulator or the tool may hit the programming pendant left unattended, which may result in personal injury and/or equipment damage.

Definition of Terms Used Often in This Manual

The MOTOMAN is the YASKAWA industrial robot product.

The MOTOMAN usually consists of the manipulator, the YRC1000micro controller, manipulator cables, the YRC1000micro programming pendant (optional), and the YRC1000micro programming pendant dummy connector (optional).

In this manual, the equipment is designated as follows:

Equipment	Manual Designation
YRC1000micro controller	YRC1000micro
YRC1000micro programming pendant	Programming pendant (optional)
Cable between the manipulator and the controller	Manipulator cable
YRC1000micro programming pendant dummy connector	Programming pendant dummy connector (optional)

Descriptions of the programming pendant keys, buttons, and displays are shown as follows:

Equipment		Manual Designation
Programming Pendant	Character Keys /Symbol Keys	The keys which have characters or symbols printed on them are denoted with [ ]. ex. [ENTER]
	Axis Keys /Number Keys	[Axis Key] and [Numeric Key] are generic names for the keys for axis operation and number input.
	Keys pressed simultaneously	When two keys are to be pressed simultaneously, the keys are shown with a “+” sign between them, ex. [SHIFT]+[COORD]
	Mode Key	Three kinds of modes that can be selected by the mode key are denoted as follows: REMOTE, PLAY, or TEACH
	Button	Three buttons on the upper side of the programming pendant are denoted as follows: HOLD button START button EMERGENCY STOP button
	Displays	The menu displayed in the programming pendant is denoted with { }. e.g. {JOB}
PC Keyboard		The name of the key is denoted. e.g. Ctrl key on the keyboard

Description of the Operation Procedure

In the explanation of the operation procedure, the expression "Select • • • " means that the cursor is moved to the object item and [SELECT] is pressed, or that the item is directly selected by touching the screen.

Registered Trademark

In this manual, names of companies, corporations, or products are trademarks, registered trademarks, or brand names for each company or corporation. The indications of (R) and TM are omitted.

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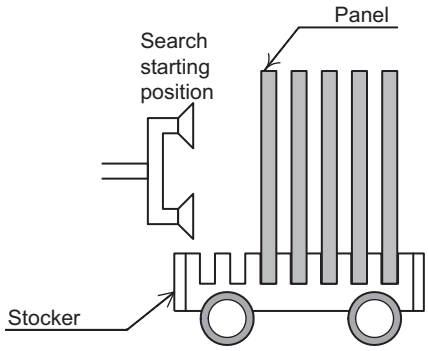
1 Search Function

The search function moves the manipulator to a specified position and stops it when a specified direct input signal comes ON.

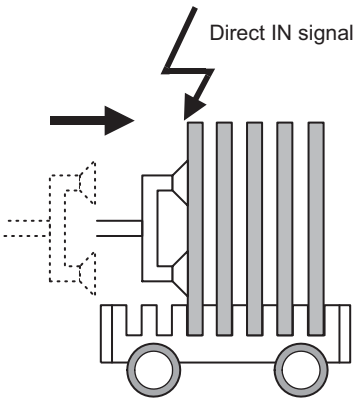
Position data at the point the manipulator stops can be taken-in. This information can be used with other instructions, for instance, to obtain the distance from a goal position and modify operation accordingly.

<Example> A manipulator performs a handling operation of panels

1. The manipulator moves to the search starting position.



2. The manipulator moves to the goal position in the search operation at low-speed. When the manipulator comes to the position to take the panel, a input signal (Direct-IN signal) from the sensor comes ON, then the manipulator stops. At this moment, the difference between the search starting position and the detected position is calculated.



3. On the base of the calculated difference, the program is modified to proceed the operation.

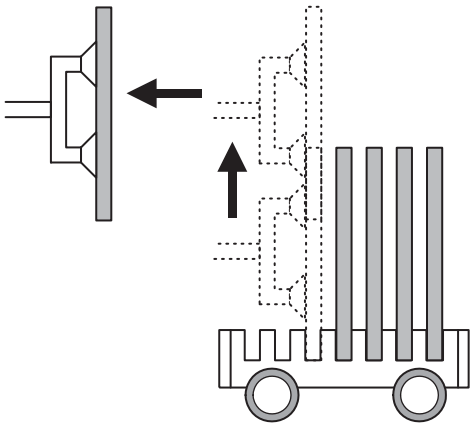


Table 1-1: JOB:WORK

Line NO.	Instruction	Content
000	NOP	
001	MOVJ VJ=20.00	
002	MOVL V=300.0 PL=0	
003	REFP 1	Set the target point of the search operation
004	CALL JOB:SRCH	Search operation
005	SFTON P002	parallel shift start
006	DOUT OT#(1)=ON	Execution of operation
007	MOVL V=100.0	
008	SFTOF	parallel shift end
009	MOVJ VJ=20.00	
010	END	

Table 1-2: JOB:SRCH

Line NO.	Instruction	Content
000	NOP	
001	GETS LPX000 $PX021	Load the target point of the search operation
002	MOVL LP000 V=100.0 SRCH RIN#(1)=ON DIS=0.1	Search operation
003	GETS LB000 $B002	Load the detection result
004	JUMP *NG IF LB000=0	Determination of the detection result
005	GETS LPX001 $PX002	Load the detected position
006	MSHIFT PX002 RF LPX000 LPX001	
007	SETE P002 (4) 0	Shift amount calculation
008	SETE P002 (5) 0	
009	SETE P002 (6) 0	
010	GETE LD000 P002 (1)	
011	GETE LD001 P002 (2)	
012	GETE LD002 P002 (3)	Shift length calculation
013	SET LR000 EXPRESS LD000 * LD000 + LD001 * LD001 + LD002 * LD002	
014	SQRT LR000 LR000	
015	JUMP *NG IF LR000>400000	Shift length check
016	JUMP *NG IF LR000<300000	
017	RET	Normal termination
018	*NG	
019	PAUSE	Abnormal termination
020	JUMP *NG	
021	END	

2 Wiring

Output signals of the sensor used in the system are connected to the YRC1000micro input port for direct-input signals. The signals that are input from this port are called direct-in signals.

The output signals of the sensor used in the system can be connected via a sequencer to the direct-in signal input port, however, processing variations may result due to scanning timing of the sequencer.

It is therefore recommended to connect the output signal of the sensor directly to the YRC1000micro input port.

2.1 Direct IN Connection

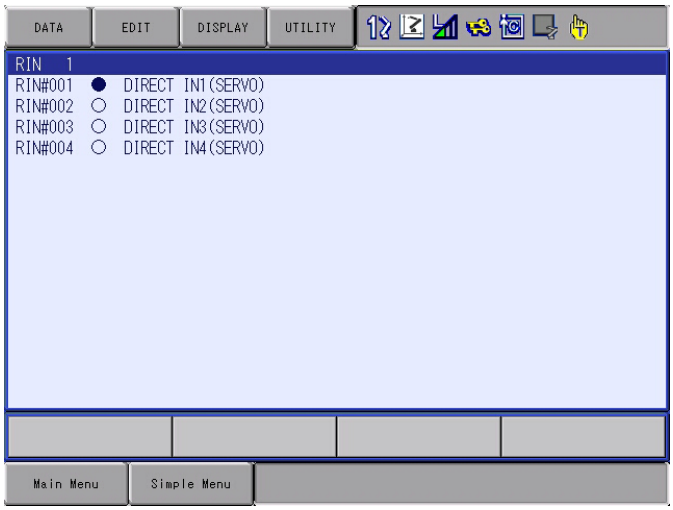
For the connection for direct-in, refer to "Chap.14.1.2 Connection of the General-Purpose I/O" in "YRC1000micro INSTRUCTIONS (RE-CTO-A222)".

2.2 Confirmation of Direct IN Signal Status

Check the status of direct-input signal on the RIN INPUT window.

2.2.1 RIN INPUT Window

1. Select {IN/OUT} under the main menu
2. Select {RIN}
   – The RIN INPUT window appears.
   – "●" indicates the ON status of direct IN signal.
   – "○" indicates the OFF status of direct IN signal



3 Instructions

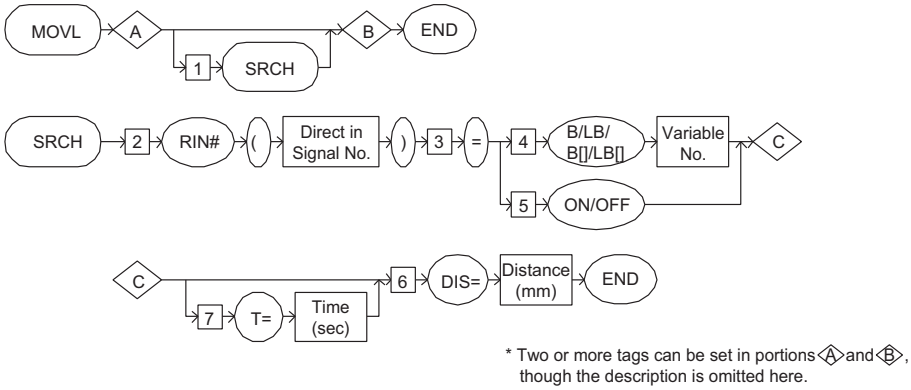
3.1 SRCH (Search Instruction)

3.1.1 Function

The SRCH is an instruction to execute the search function. It is set as an additional item to the move instruction.

- MOVL
- IMOV
- SMOVL

3.1.2 Construction

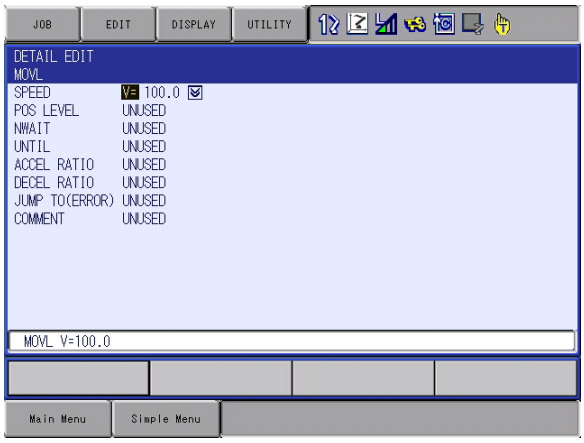


3.1.3 Explanation

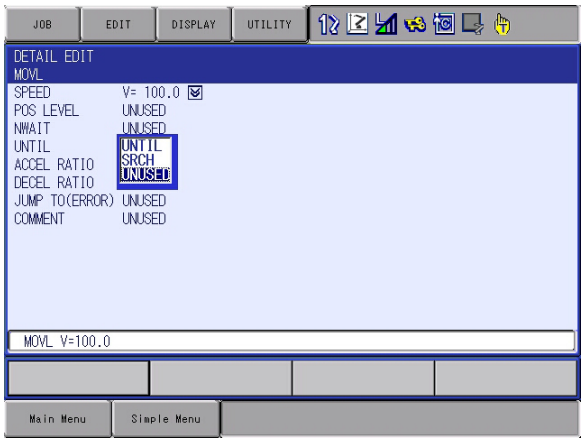
№	Tag	Explanation	Note
1	SRCH	Specifies the SRCH instruction. SRCH instruction is an instruction to execute search operation.	
2	RIN# (Input number)	Specify direct in signal number.	No.:1 - 4 B/I/D/B[ ]/I[ ]/D[ ]/LB/LI/LD/LB[ ]/LI[ ]/LD[ ] can be used.
3	=	It is equal.	
4	B Variable number/ LB Variable number/ B [Array number]/ LB [Array number]/	Specify byte type variables which are used as detection conditions.	Last significant bit: 0: OFF 1: ON
5	ON/OFF	Specify detection conditions by ON/OFF.	
6	Distance	Specify the distance to extend the target point.	Distance: 0.1 - 6553.5mm Variable B/I/D/B[ ]/I[ ]/D[ ]/LB/LI/LD/LB[Distance can be specified with variables B/I/D/B[]/D[]/LB/LI/LD/LB[]/LI[]/LD[] (unit: 0.1 mm).
7	Time	When using direct in signal search start delay time (in 0.001 s) and start point search unit, set 0.1 s or more. This is because there may be cases where the relay (200 V) is switched when starting search operation and direct in signal is constantly received.	unit: 0.001 - 65.535 s Time can be specified with variables I/LI/I[]/LI[] (unit: 0.001 s).

3.1.4 Register SRCH Instruction

1. Move the cursor to the instruction area.
2. Move the cursor to the move instruction where SRCH instruction is to be registered.
3. Press [SELECT] twice.
   (1) Press [SELECT] once to display the contents registered in the job in the input buffer line.
   (2) Press [SELECT] once more, and the DETAIL EDIT window of the move instruction appears.

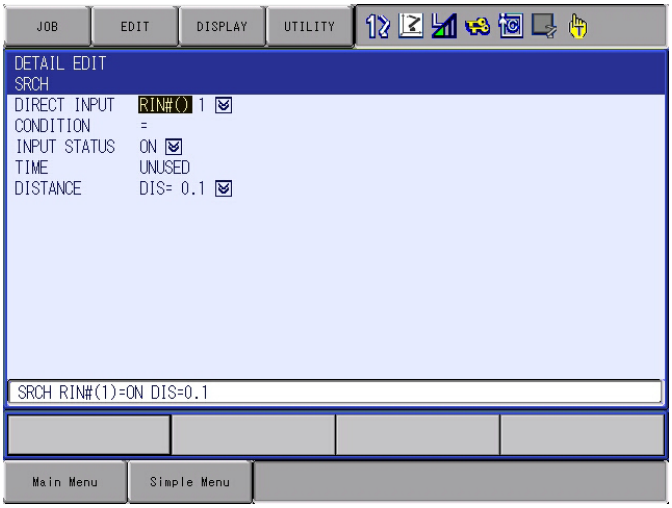


4. Select {UNTIL}.



5. Select {SRCH}.

6. Set each item on the DETAIL EDIT window of SRCH instruction.
   – The DETAIL EDIT window of SRCH instruction appears.



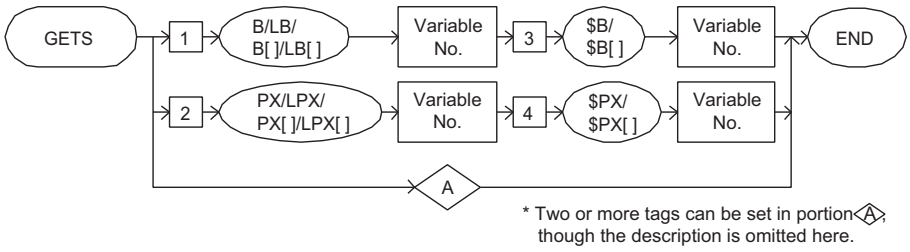
7. Press [ENTER].
   – Returns to the DETAIL EDIT window of the move instruction.
8. Press [ENTER] twice.
   (1) Press [ENTER] once to display the set contents in the input buffer line.
   (2) Press [ENTER] once more, and the set contents are registered in the job.

3.2 GETS Instruction

3.2.1 Function

Saves the system variable ($ variable) as a user variable.

3.2.2 Construction



3.2.3 Explanation

№	Tag	Explanation	Note
1	B Variable number/ LB Variable number/ B [Array number]/ LB [Array number]	Specify byte type variable number to save system variable.	
2	PX Variable number/ LPX Variable number/ PX [Array number]/ LPX [Array number]	Specify expanded position type variable number to save system variable.	
3	$B Variable number/ $B [Array number]	Specify byte type system variable number to set.	
4	$PX Variable number/ $PX [Array number]	Specify expanded position type variable number to set.	

SUPPLE -MENT

System variable

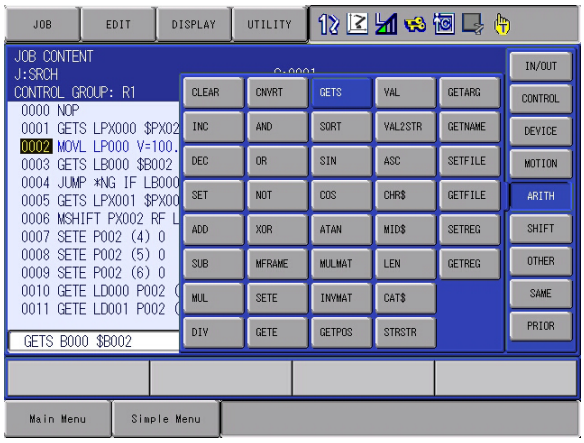
The GETS is the only instruction that refers to system variables written by the controller system.

The following system variables are available.

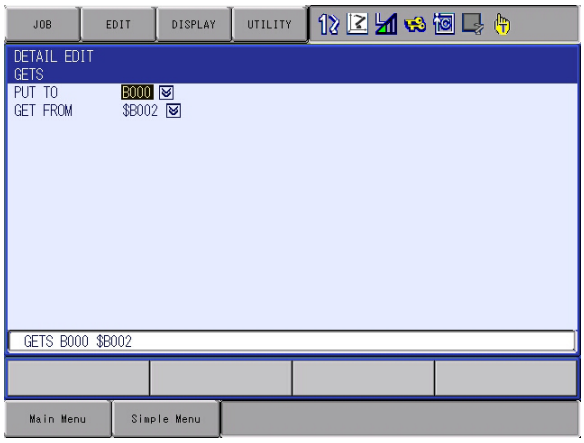
System Variable	Type	No.	Explanation
$B type variable	Byte type	$B002	Specifies detected/not detected of the SRCH instruction 0: Not detected, 1: Detected
$PX type variable	Expanded position type	$PX002	Position detected by the optional SRCH instruction (pulse type)
		$PX003	Position detected by the optional SRCH instruction (XYZ type)
		$PX011 - 018	Teaching position by REFP 1 - 8 instruction (pulse type)
		$PX021 - 028	Teaching position by SREFP 1 - 8 instruction (pulse type)

3.2.4 Register GETS Instruction

1. Move the cursor to the address area.
2. Move the cursor to the line where GETS instruction is to be registered.
3. Press [INFORM LIST].
   (1) The instruction list dialog box appears.
   (2) The cursor moves to the instruction list dialog box while the cursor in the address area changes to an under bar.
4. Select {GETS}.
   – At this stage, the instruction on the cursor position is displayed with the previously registered additional items in the input buffer line.



5. Press [SELECT] twice.



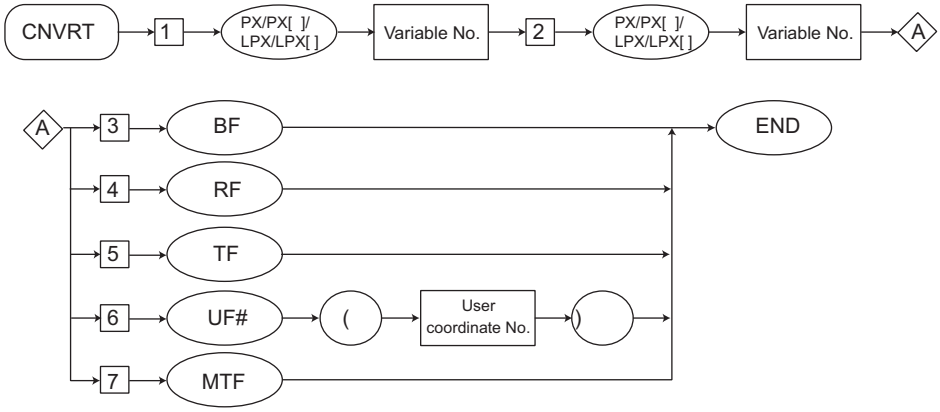
6. The DETAIL EDIT window of GETS instruction appears.
7. Press [ENTER] twice.
   (1) Press [ENTER] once to display the set contents in the input buffer line.
   (2) Press [ENTER] once more, and the set contents are registered in the job.

3.3 CNVRT (Position Type Variable Conversion Instruction)

3.3.1 Function

The CNVRT is an instruction to convert a pulse data position type variable into XYZ type position type variable of the specified coordinate system.

3.3.2 Construction



3.3.3 Explanation

№	Tag	Explanation	Note
1	PX Variable number/ LPX Variable number/ PX [Array number]/ LPX [Array number]	Specifies the number of the expanded position type variable where the converted data is stored.	
2	PX Variable number/ LPX Variable number/ PX [Array number]/ LPX [Array number]	Specifies the number of the expanded position type variable to be converted.	
3	BF	Specifies the conversion in the base coordinate system.	
4	RF	Specifies the conversion in the robot coordinate system.	
5	TF	Specifies the conversion in the tool coordinate system.	
6	UF# (User coordinate number)	Specifies the conversion in the user coordinate system.	No.: 1 - 63 Variable B/I/D/LB/LI/LD can be used.
7	MTF	Specifies the conversion on the master tool coordinate system. On the master tool coordinate system, the data is converted to a position relative to the master manipulator.	Available only with the optional independent coordinate function.

SUPPLE -MENT

Expanded position type variables

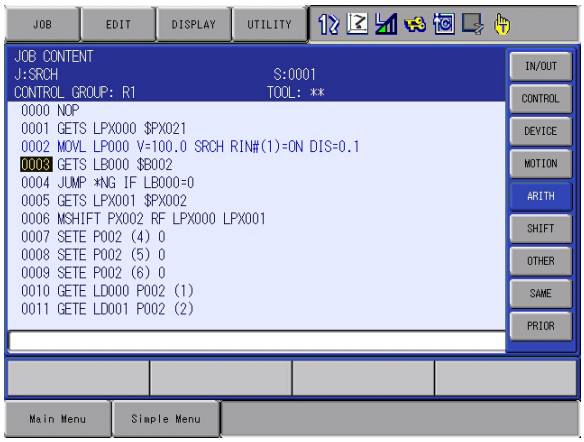
The expanded position type variable is a position type variable that depends on the control group in the job.

<Example>

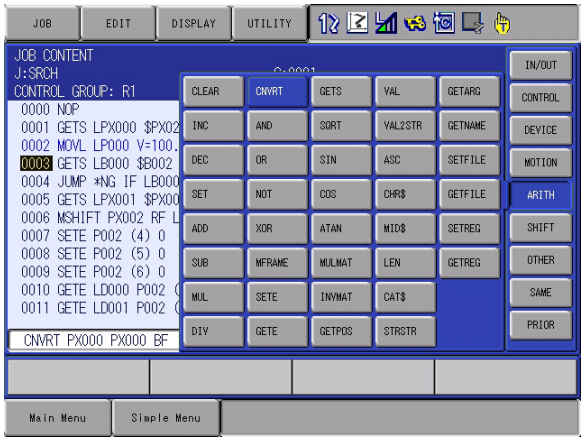
- When the control group is R1:
  PX000 indicates P000.
- When the control group is R1 + B1:
  PX000 indicates P000 and BP000.
- When the control group is R1 + B1 + ST1:
  PX000 indicates P000 + BP000 + EX000.
- When the control group is R1 + R2 + B1 + B2 + ST1 in the coordinated job (master R1 + B1):
  PX000 indicates the following:
  P000: R2 (slave)
  P001: R1 (master)
  BP000: B2 (slave)
  BP001: B1 (master)
  EX000: ST1

3.3.4 CNVRT Instruction

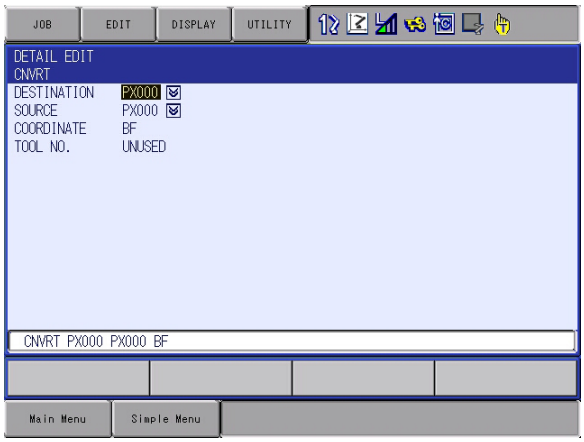
1. Move the cursor to the address area.
2. Move the cursor to the line where CNVRT instruction is to be registered.
3. Press [INFORM LIST].
   (1) The instruction list dialog box appears.
   (2) The cursor moves to the instruction list dialog box while the cursor in the address area changes to an under bar.



4. Select {CNVRT}.
   – At this stage, the instruction on the cursor position is displayed with the previously registered additional items in the input buffer line.



5. Press [SELECT] twice.
   – The DETAIL EDIT window of CNVRT instruction appears.



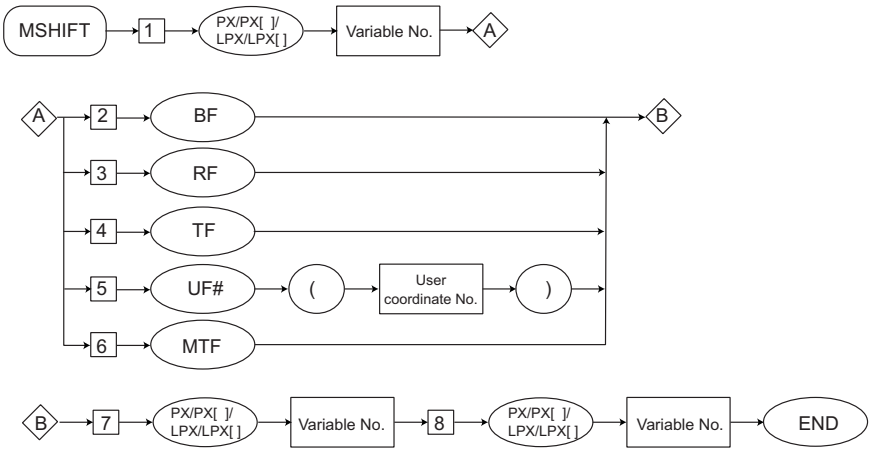
6. Set variables on the DETAIL EDIT window of CNVRT instruction
7. Press [ENTER] twice.
   (1) Press [ENTER] once to display the set contents in the input buffer line.
   (2) Press [ENTER] once more, and the set contents are registered in the job.

3.4 MSHIFT Instruction

3.4.1 Function

Calculate shift amount in the specified coordinate system from the reference position and the target position.

3.4.2 Construction



3.4.3 Explanation

1. PX Variable number/LPX Variable number/PX [array number]/LPX [array number]
   – Add the following tag.

№	Tag	Explanation	Note
1	PX Variable number/ LPX Variable number/ PX [Array number]/ LPX [Array number]	Specifies the number of the expanded position variable to store the calculated shift.	<Data 1>

2. BF/RF/TF/UF# (User coordinate number)/MTF

№	Tag	Explanation	Note
2	BF	Specifies the conversion in the base coordinate system.	
3	RF	Specifies the conversion in the robot coordinate system.	
4	TF	Specifies the conversion in the tool coordinate system.	
5	UF# (User coordinate number)	Specifies the conversion in the user coordinate system.	No.: 1 - 63 Variable B/I/D/LB/LI/LD can be used.
6	MTF	Specifies the conversion on the master tool coordinate system. On the master tool coordinate system, the data is converted to a position relative to the master manipulator.	Available only with the optional independent/coordinated function.

3. PX Variable/LPX Variable/PX [array number]/LPX [array number]

№	Tag	Explanation	Note
7	PX Variable number/ LPX Variable number/ PX [Array number]/ LPX [Array number]	Specifies the number of the expanded position variable to store the calculated shift.	<Data 2>

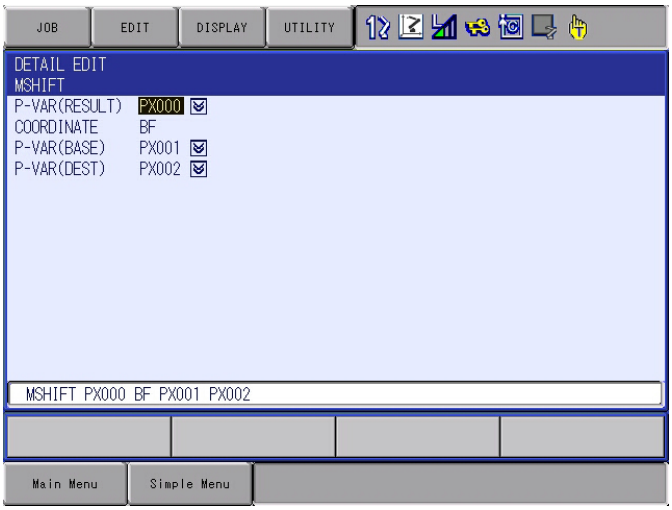
4. PX Variable number/LPX Variable number/PX [array number]/LPX [array number]

№	Tag	Explanation	Note
8	PX Variable number/ LPX Variable number/ PX [Array number]/ LPX [Array number]	Specifies the expanded position type variable number of the target position to calculate the amount of the shift.	<Data 3>

3.4.4 Register MSHIFT Instruction

1. Move the cursor to the address area.
2. Move the cursor to the line where MSHIFT instruction is to be registered.
3. Press [INFORM LIST].
   (1) The instruction list dialog box appears.
   (2) The cursor moves to the instruction list dialog box while the cursor in the address area changes to an under bar.
4. Select {MSHIFT}.
   – At this stage, the instruction on the cursor position is displayed with the previously registered additional items in the input buffer line.

5. Press [ENTER] twice.
   – The DETAIL EDIT window of MSHIFT instruction windows appears.



6. Set variables on the DETAIL EDIT window of MSHIFT instruction.
7. Press [ENTER] twice.
   (1) Press [ENTER] once to display the set contents in the input buffer line.
   (2) Press [ENTER] once more, and the set contents are registered in the job.

4 Alarm List

Alarm No.	Message	Cause	Remedy
4474	WRONG CONTROL GROUP AXIS	The CALL/JUMP/PSTART destination job could not be executed. An attempt was made to call or jump to a job whose control group cannot be controlled. (Subcode: The related control-group)	(1) Confirm the settings below. • Make the setting in advance so that the control group of the CALL/JUMP destination job is included in that of the CALL/JUMP source job. • Don't start a job containing a running control group with a "PSTART" instruction.
4499	UNDEFINED POSITION VARIABLE	The position type variable is not registered. An attempt was made to use the position type variable that was not set. (Subcode: The variable number)	(1) Confirm the settings below. Set the position type variable.
4507	REFP POS ERROR (SEARCH MOTION)	Incorrect teaching point for search detection The search start point and the motion target point are the same, or the distance between the two points is too short.	(1) Confirm the settings below. • Perform the teaching again so that the search start point and the motion target point are not the same. • Increase the distance between the search start point and the motion target point.

5 Instruction List

- < > indicates numerical or alphabetical data.
- If multiple items are shown in one section, select one of the items.

SRCH	Function	Executes a search function.		
	Additional Item	RIN#(<direct IN No.>) = <status>		Direct IN No. : 1 - 4 Status : ON, OFF, B <variable No.>
		T = <time·s>		Direct IN signal input check start delay time
		DIS = <distance·mm>		A length of passing over the target point specified by the position type variable
	Example	MOVL P000 V=138 SRCH RIN#(1)=ON T=1.00 DIS=10.0		
GETS	Function	Saves the system variable as a user variable.		
	Additional Item	B <variable No.>, I <variable No.>, D <variable No.>, R <variable No.>, PX <variable No.>		User variable
		$B <variable No.>, $I <variable No.>, $D <variable No.>, $R <variable No.>, $PX <variable No.>		System variable
	Example	GETS B000 $B000 GETS I001 $I[1] GETS PX003 $PX001		
CNVRT	Function	Converts a pulse data position type data of Data 2 into a Cartesian data position type variable using a specified coordinate system, then saves in Data 1. Format : CNVRT <Data 1><Data 2> coordinate system		
	Additional Item	Data 1	PX <variable No.>	
		Data 2	PX <variable No.>	
		BF, RF, TF, UF# (<user coordinate system No.>), MTF		BF:Base coordinate system RF : Robot coordinate system TF : Tool coordinate system UF : User coordinate system MTF : Master tool coordinate system
	Example	CNVRT PX000 PX001 BF CNVRT LPX000 LPX001 TF		

SEFTOF	Function	Stop shift operation.		
	Additional Item	No		
	Example	SFTOF		
MSHIFT	Function	Calculate shift amount in the specified coordinate system from data 2 and data 3 and store the calculated value in data 1. Format : MSHIFT <Data 1><Coordinates><Data 2><Data 3>		
	Additional Item	Data 1	PX <variable No.>	
		Coordinate	BF, RF, TF, UF# (<user coordinate system No.>), MTF	BF : Base coordinate system RF : Robot coordinate system TF : Tool coordinate system UF : User coordinate system MTF : Master tool coordinate system
		Data 2	PX <variable No.>	
		Data 3	PX <variable No.>	
	Example	MSHIFT PX100 RF PX001 PX002		

YRC1000micro OPTIONS
INSTRUCTIONS

FOR SEARCH FUNCTION

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Specifications are subject to change without notice
for ongoing product modifications and improvements.

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YASKAWA ELECTRIC CORPORATION

© Printed in Japan July 2017 17-07

MANUAL NO.
HW1484472